

ORLANDO E. MARTÍNEZ-DURIVE

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PROFILES:  [INSTITUTIONAL WEBSITE](#) |  [GOOGLE SCHOLAR](#) |  [LINKEDIN](#)

I am a data scientist and machine learning researcher with 8+ years of experience at the intersection of AI, networked systems, and critical infrastructure. My work combines developing ML models, building production systems, and solving real-world problems—from optimizing mobile network energy efficiency to analyzing political polarization through app consumption patterns. I have worked across academia and industry, including research internships at Telefónica Innovación Digital, Cisco ThousandEyes, and the Network Science Institute at Northeastern University.

EDUCATION

- 2021 – 2025 **PhD in Telematics Engineering**, University Carlos III of Madrid, Spain.
Graduated **Cum Laude** with International Doctorate distinction.
Thesis: Leveraging Mobile Network Metadata for Understanding Societal Phenomena and Optimizing Network Operations.
- 2019 – 2021 **MSc in Computer Science**, University of Havana, Cuba.
Thesis: A System to Analyze Human Mobility in Cuba Using Mobile Phone Data.
- 2012 – 2017 **BSc in Computer Science**, University of Havana, Cuba.
Thesis: Exploring Low Auto-Correlation Binary Sequences Using Message-Passing Algorithms.

PROFESSIONAL & RESEARCH EXPERIENCE

- 2025 – Present **NetAI, Edinburgh, Scotland | Machine Learning Researcher**
Developing AI/ML models for energy-efficient network operation, focusing on automated detection of underutilized resources and intelligent RAN control policies.
- 2025 – Present **IMDEA Networks Institute, Madrid, Spain | Postdoctoral Researcher**
Researching next-generation mobile architectures and interpretable AI/ML systems for reliable and sustainable 5G and B5G networks.
- 2025 **Cisco ThousandEyes, Crete, Greece | Research Intern**
Analyzed Internet routing to assess outage risks related to the BGP Maximum-Prefix limit.
- 2024 **Network Science Institute, Boston, MA, United States | Visiting Researcher**
Examined mobile app consumption and political polarization using large-scale traffic data.
- 2023 – 2024 **Telefónica Innovación Digital, Madrid, Spain | Visiting Researcher**
Measured the impact of energy-saving and carbon-reduction policies in mobile networks, studying trade-offs between energy efficiency and user experience.
- 2020 – 2025 **IMDEA Networks Institute, Madrid, Spain | Research Assistant**
Developing new methods to improve spatial accuracy, enhance 5G adoption, increase accessibility of mobile traffic metadata, optimize energy-saving strategies, and analyze political orientation.
- 2019 **Politecnico di Torino, Turin, Italy | Visiting Researcher**
Developed a visualization tool for human mobility flows obtained from Call Detail Records (CDRs).
- 2017 – 2020 **Physics Department, University of Havana, Havana, Cuba | Research & Teaching Assistant**
Developed and maintained BDPHONEFLOW, software for human mobility analysis based on CDRs.
- 2019 – 2020 **Cubyke, Havana, Cuba | Full-Stack Developer and Web Analyst**
Developed and maintained the company website. Integrated third-party payment systems via API. SEO enhancement using Google Search Console and AdSense. Created data-driven user behavior analytics reports.
- 2017 – 2020 **Taxi Drivers Cuba, Havana, Cuba | Full-Stack Developer and Web Analyst**
Developed and maintained the company website. Implemented an online booking system and integrated PayPal API. SEO enhancement and structured product data implementation.

TECHNICAL SKILLS

Programming: Python, R, C++, Bash, JavaScript

Geospatial and Mobility Analysis: CDR/XDR metadata, GPS data, Voronoi mapping, antenna coverage estimation, clustering, OpenStreetMap API, Google Maps API.

Software & Web Development: GitHub, GitLab, CI/CD, APIs, Webpack, Apache, HTML, CSS, Google Analytics, Google Ads, PHP, SQL.

Big Data & Cloud Computing: Apache Spark, Hadoop, GeoPandas, NumPy, Amazon Web Services (AWS), Google Cloud Platform (GCP), Jupyter, Google Colab.

Machine Learning: TensorFlow, PyTorch, scikit-learn, Generative Adversarial Networks (GANs), Graph Neural Networks (GNNs), Transformers.

Networking: Cellular network analysis (4G/LTE, 5G/NR), BGP routing (BGPPKit, BGPReader), Sionna RT.

Data Visualization: Matplotlib, Plotly, D3.js, Folium, Mapbox, Flowmap.blue.

TEACHING

- 2023 – 2024 **University Carlos III of Madrid, Spain | Teaching Assistant.**
Supported ≈ 50 undergraduate students in Telematics Engineering and Sound and Image Engineering in practical sessions for **Access Networks and Shared Media**, using Python and mathematical models.
- 2018 - 2020 **University of Havana, Cuba | Teaching Assistant.**
- **Computational Methods II.** Supported lab sessions for ≈ 20 undergraduate Physics students. Using C++, Python, and NumPy.
 - **Probability and Statistics.** Provided support and feedback for ≈ 20 undergraduate Physics students in practical sessions. Using simulation tools in Python and Excel.
 - **Algebra II.** Supported ≈ 40 undergraduate Physics students in practical sessions.
 - **Numerical Methods.** Guided ≈ 30 undergraduate Physics students in practical sessions. Using MATLAB, Python, and C++.
 - **Mathematical Analysis I.** Provided support and feedback to ≈ 40 undergraduate Physics students in practical sessions.
 - **Computational Methods I.** Supported lab sessions of ≈ 50 undergraduate Physics students. Using Linux, terminal scripting, and Ubuntu.
 - **Algebra I.** Supported ≈ 60 undergraduate Computer Science students in practical sessions.

SUPERVISED STUDENTS

- 2023 - 2024 Daniel Amaro, **M.Sc. student at the University of Havana, Cuba.**
Assisted and supported Daniel in his project, *Trajectory Reconstruction from Mobile Traffic Metadata*.
- 2023 Prashant Ray, **M.Sc. student at Politecnico di Torino, Italy.**
Collaborated with Prashant in his M.Sc. thesis project, *Land Use Detection through Matrix Decomposition*.
- 2020 - 2021 Theo Couturier, **Undergraduate student at l'ENSEEIHPT HPC, France.**
Collaborated with Theo on a research project, *Beyond Voronoi: Plain Probabilistic Spatial Coverage Inference from Base Station Deployments*.
- 2019 Roger Casimiro, **Undergraduate student at Technological University José Antonio Echeverría, Cuba.**
Provided support and feedback to Roger in his research project, *"Human Mobility Using Mobile Phone Records."*

COMMUNITY SERVICE

Technical Program Committee (TPC) Member

- Agu. 2026 IEEE INFOCOM, 2027.
- Oct. 2025 Passive and Active Measurement Conference (PAM), 2025, 2026.
- Aug. 2025 Shadow TPC, ACM Internet Measurement Conference (IMC), 2022, 2025 and 2026.
- Feb. 2024 1st IEEE Workshop on Quality of Data in Network Telemetry, Co-located with the IEEE/IFIP Network Operations and Management Symposium.
- Sept. 2023 NetMob Data Challenge, Co-located with the NetMob Conference.

External Reviewer

- 2025 IEEE Networking Letters.
- 2024 IEEE Transactions on Network and Service Management (IEEE TNSM).

2022	International Symposium on Personal, Indoor, and Mobile Radio Communications (PIMRC).
2022	Network Traffic Measurement and Analysis Conference (TMA).
2021, 2022	International Conference on Networking.
2022	Journal of Maps (JoM).
2022	Sustainable Cities and Society (SCS).
2021	Complex Networks.

Conference Organization

May 2023	Web Chair & Local Organizer, NetMob 2023 Conference and Data Challenge.
Sept. 2022	Web Chair & Local Organizer, Passive and Active Measurement Conference (PAM) 2023.

Science Fair Exhibitor

March 2025	Can We Play With Our Head? Using machine learning and computer vision tools to control a rocket with head movements. Madrid Science Fair (IFEMA, Madrid, Spain); Edition XIV .
March 2024	How Do Computers See the World and Applications? Demonstrated machine learning using Google MediaPipe Pose Landmarker, OpenCV, and PyGame to recognize faces, apply filters, and enable gesture-based human-computer interaction. Madrid Science Fair (IFEMA, Madrid, Spain); Edition XIII .
March 2023	How Does the Internet Work and Reach Our Homes? Exhibited and explained the computer network hardware that powers the Internet and its access methods. Madrid Science Fair (IFEMA, Madrid, Spain); Edition XII .

PUBLICATIONS

May 2025	Martínez-Durive, O. E. , Suárez-Varela, J., Omaña-Iglesias, J., Lutu, A., & Fiore, M. "An Evaluation of RAN Sustainability Strategies in Production Networks." IEEE Conference on Computer Communications (INFOCOM), London, United Kingdom. May 2025. DOI: 10.1109/INFOCOM55648.2025.11044549 .
Dec. 2024	Martínez-Durive, O. E. , Bakirtzis, S., Ziemlicki, C., Zhang, J., Wassell, I. J., & Fiore, M. "DeepMEND: Reliable and Scalable Network Metadata Geolocation from Base Station Positions." IEEE International Conference on Sensing, Communication, and Networking (SECON), Phoenix, AZ, United States. Dec. 2024. DOI: SECON64284.2024.10934920 .
Nov. 2024	Zanella, A., Madariaga, D., Mishra, S., Martínez-Durive, O. E. , & Fiore, M. "Characterizing, Modeling, and Exploiting the Mobile Demand Footprint of Large Public Protests." ACM Internet Measurement Conference (IMC), Madrid, Spain, Nov. 2024. DOI: 10.1145/3646547.3688421 .
Oct. 2024	Martínez-Durive, O. E. , Madariaga, D., Ray, P., Amaro-Ramos, D. A., & Fiore, M. "Uncovering Latent Patterns in Service-Level Spatiotemporal Mobile Traffic." ACM SIGSPATIAL International Workshop on Geopriacy and Data Utility for Smart Societies, Atlanta, GA, United States. Oct. 2024. DOI: 10.1145/3681768.3698506 .
May 2024	Mishra, S., Zanella, A., Martínez-Durive, O. E. , Madariaga, D., & Fiore, M. "Characterizing 5G Adoption and Its Impact on Network Traffic and Mobile Service Consumption." IEEE Conference on Computer Communications (INFOCOM), Vancouver, Canada. May 2024. DOI: 10.1109/INFOCOM52122.2024.10621344 .
May 2024	Martínez-Durive, O. E. , Úcar, I., Smoreda, Z., Moro-Egido, E., & Fiore, M. "Mobile App Consumption and Political Orientation." IEEE Conference on Computer Communications Workshops (INFOCOM WKSHPS), Vancouver, Canada. May 2024. DOI: 10.1109/INFOCOMWKSHPS61880.2024.10620873 .
May 2024	Zanella, A., Martínez-Durive, O. E. , Mishra, S., Madariaga, D., Ziemlicki, C., & Fiore, M. "Impact of Public Protests on Mobile Networks." IEEE Conference on Computer Communications Workshops (INFOCOM WKSHPS), Vancouver, Canada. May 2024. DOI: 10.1109/INFOCOMWKSHPS61880.2024.10620747 .
Jun. 2023	Martínez-Durive, O. E. , Mishra, S., Ziemlicki, C., Rubrichi, S., Smoreda, Z., & Fiore, M. "France Through the Lens of Mobile Traffic Data." 7th Network Traffic Measurement and Analysis Conference (TMA), Naples, Italy. Jun. 2023. DOI: 10.23919/TMA58422.2023.10198971 .
May 2023	Martínez-Durive, O. E. , Mishra, S., Ziemlicki, C., Rubrichi, S., Smoreda, Z., & Fiore, M. "The NetMob23 Dataset: A High-resolution Multi-region Service-level Mobile Data Traffic Cartography." Preprint arXiv. May 2023. DOI: arXiv.2305.06933 .
Sept. 2022	Martínez-Durive, O. E. , Couturieux, T., Ziemlicki, C., & Fiore, M. "VoronoiBoost: Data-driven Probabilistic Spatial Mapping of Mobile Network Metadata." IEEE International Conference on Sensing, Communication, and Networking (SECON), Stockholm, Sweden. Sept. 2022. DOI: 10.1109/SECON55815.2022.9918610 .

- May 2022 Zarella, A., **Martínez-Durive, O. E.**, Mishra, S., Smoreda, Z., & Fiore, M. "Impact of Later-Stage COVID-19 Response Measures on Spatiotemporal Mobile Service Usage." IEEE Conference on Computer Communications (INFOCOM), London, United Kingdom. May 2022. DOI: [10.1109/INFOCOM48880.2022.9796888](https://doi.org/10.1109/INFOCOM48880.2022.9796888).
- Oct. 2021 García-Borroto, M., **Martínez-Durive, O. E.**, Sánchez, E., Díaz, H., & Lage-Castellanos, A. "Intra-day Population Fluxes from Mobile Phone Data in Havana, Cuba." 7th International Workshop on Artificial Intelligence and Pattern Recognition (IWAIPR), Havana, Cuba. Oct. 2021. DOI: [10.1007/978-3-030-89691-1_37](https://doi.org/10.1007/978-3-030-89691-1_37).
- July 2021 **Martínez-Durive, O. E.**, Kotsireas, I. S., Mulet-Genicio, R., & Lage-Castellanos, A. "Low Auto-correlation Binary Sequences Explored Using Warning Propagation." Revista Cubana de Física, 38(1). Jul. 2021. ISSN: 0253-9268. [Online ISSN 2224-7939](https://doi.org/10.1007/978-3-030-89691-1_37).
- Apr. 2020 Lage-Castellanos, A., & **Martínez-Durive, O. E.** "Evolución de la movilidad poblacional durante 100 días de COVID-19 en Cuba: un análisis a partir de los datos de la telefonía móvil." Editorial UH, 2020. ISBN: [978-959-7251-89-7](https://doi.org/10.1007/978-3-030-89691-1_37).
- Apr. 2020 Remond Noa, R., Pérez Rodríguez, N., Torres Reyes, A., Veranes Miranda A., Fernández Lorenzo, J. M., Oviedo Álvarez, V., De la Garma, D., Lage Castellanos, A., **Martínez-Durive, O. E.**, Más Bermejo, P., Sánchez Valdés, L., Vidal Ledo, M. J., & Almeida Cruz, Y. "Análisis, mapas y modelos para el seguimiento de la pandemia"; La Habana: atlas de la COVID-19, Editorial UH, 2020. ISBN-e: [978-959-7251-91-0](https://doi.org/10.1007/978-3-030-89691-1_37).

PRESENTATIONS

- April. 2026 **What happens when you overshare? A Look into the BGP Maximum-Prefix Limit.**
SEE 14. Belgrade, Serbia. April. 2026.
- Oct. 2025 **What happens when you overshare? A Look into the BGP Maximum-Prefix Limit.**
Routing Working Group, RIPE 91. Bucharest, Romania. Oct. 2025.
- Oct. 2024 **Mobile App Consumption and Its Relationship with Polarization.**
8th edition of the NetMob Conference, World Bank, Washington, D.C., United States. Oct 2024.
- Oct. 2024 **Characterizing 5G Adoption and Its Impact on Network Traffic and Mobile Service Consumption.**
8th edition of the NetMob Conference, World Bank, Washington, D.C., United States. Oct 2024.
- Oct. 2023 **Mobile App Consumption and Political Orientation.**
7th edition of the NetMob Conference, Universidad Carlos III de Madrid, Madrid, Spain. Oct. 2023.
- Oct. 2023 **Simplifier: Scalable Spatial Diffusion of Mobile Network Metadata from Base Station Locations.**
7th edition of the NetMob Conference, Universidad Carlos III de Madrid, Madrid, Spain. Oct. 2023
- Dec. 2022 **Spatial Coverage Inference from Base Station Deployments.**
Seminar at IMDEA Networks Institute, Madrid, Spain. Dec. 2022.
- March 2022 **Beyond Voronoi: Plain Probabilistic Spatial Coverage Inference from Base Station Deployments.**
Seminar at IMDEA Networks Institute, Madrid, Spain. March 2022.
- Nov. 2022 **Apache Spark and Hadoop in Google Cloud from Scratch.**
Training session for the Big dATA aNalytics for Radio Access Networks (BANYAN) project, IMDEA Networks Institute, Madrid, Spain. Nov. 2022.
- Feb. 2020 **Epidemic Simulation Based on Human Mobility Patterns.**
National Meeting of Young Physicists, Institute of Cybernetics, Mathematics, and Physics, Havana, Cuba. Feb. 2020.
- Nov. 2019 **Human Mobility in Cuba Using Mobile Phone Data.**
International School on Dengue, Tropical Medicine Institute Pedro Kourí (IPK), Havana, Cuba. Nov. 2019.

POSTERS & DEMOS

- July, 2025 **Martínez-Durive, O. E.**, Bakirtzis, S., Ziemlicki, C., & Fiore, M.
"Modeling Base Station Metadata Geolocation."
NetMob, Paris, France. October 2025.
- May 2025 **Martínez-Durive, O. E.**, Bakirtzis, S., Ziemlicki, C., & Fiore, M.
"Demonstrating Deep Learning-based Spatial Diffusion."
IEEE Conference on Computer Communications (INFOCOM), London, United Kingdom. May 2025.
- Feb. 2025 **Martínez-Durive, O. E.**, Suarez-Varela, J., Omaña-Iglesias, J., Lutu, A., & Fiore, M.
"An Evaluation of RAN Sustainability Strategies in Production Networks."
1st European Mobile Systems Winter School: *Hands-on AI and Mobile Systems: Research and Practice*, Como, Italy. Feb. 2025.

- Oct. 2024 Zarella, A., **Martínez-Durive, O. E.**, Mishra, S., Madariaga, D., Ziemlicki, C., & Fiore, M.
"Impact of Public Protests on Mobile Networks."
8th edition of the NetMob Conference, World Bank, Washington, D.C., United States. Oct. 2024.
- May 2024 **Martínez-Durive, O. E.**, Úcar, I., Smoreda, Z., Moro-Egido, E., & Fiore, M.
"What Can Mobile App Consumption Reveal About Election Outcomes?"
International Student Workshop *The Interplay Between Machine Learning and Communication Systems*,
Centre Tecnològic de Telecomunicacions de Catalunya (CTTC), Barcelona, Spain. May 2024.
- Jun. 2023 **Martínez-Durive, O. E.**, Bakirtzis, S., Wassell, I., Ziemlicki, C., & Fiore, M.
"Simplifier: Spatial Diffusion of Mobile Network Metadata from Base Station Locations."
7th edition of the Network Traffic Measurement and Analysis Conference, Naples, Italy. Jun. 2023.
- May 2023 **Martínez-Durive, O. E.**, Ray, P., & Fiore, M.
"Mobile Traffic Analysis Using Tensor Decomposition."
13th IMDEA Networks Annual International Workshop, Madrid, Spain. May 2023
- May 2023 **Martínez-Durive, O. E.**, Mishra, S., Ziemlicki, C., Rubrichi, S., Smoreda, Z., & Fiore, M.
"The NetMob23 Dataset: A High-resolution Multi-region Service-level Mobile Data Traffic Cartography."
13th IMDEA Networks Annual International Workshop, Madrid, Spain. May 2023.
- Jun. 2022 **Martínez-Durive, O. E.**, Couturieux, T., Ziemlicki, C., & Fiore, M.
"Beyond Voronoi: Plain Probabilistic Spatial Coverage Inference from Base Station Deployments."
12th IMDEA Networks Annual International Workshop, Madrid, Spain. Jun. 2022

RESEARCH PROJECTS

ORIGAMI: Optimized resource integration and global architecture for mobile infrastructure for 6G.

Funded by the European Union, Grant No. [101139270](#).

NetSense: Network Sensing.

Funded by the Regional Government of Madrid, Grant No. [2023-5A/TIC-28944](#) and [2019-T1/TIC-16037](#).

Mobility Data for Epidemiologists Based on Cell Phones.

Funded by the Pan American Health Organization (PAHO/WHO), [IS4H projects](#).

REFERENCES

Prof. Marco Fiore – PhD Supervisor

Research Professor at IMDEA Networks, Madrid, Spain.

Chief Technology Officer (CTO) at Net AI, Edinburgh, United Kingdom.

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Antonis Chariton – Supervisor at Cisco ThousandEyes

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Prof. Esteban Moro Egido – Supervisor at The Network Science Institute

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